DECK 20 · WEEK 4 · AFTERNOON 5

Portfolio Management, MPT, CAPM & Risk

How the market prices risk — and how investors build portfolios to harvest the reward for bearing it

CFA Level I · Vol 9 Learning Modules 1–6 · FINAL DECK (20 of 20)

Roadmap — Portfolio Management, MPT, CAPM & Risk Management

6 sections · ~2 hours · Built bottom-up from CFA readings

01	**MPT foundations**	risk aversion & utility · two-asset variance · correlation & diversification · efficient frontier · min-variance portfolios
02	**CAL, CML, CAPM & SML**	capital allocation line · market portfolio · systematic vs idiosyncratic · β · CAPM · security market line · under/overpriced assets
03	**The portfolio management process**	planning · execution · feedback · individual vs institutional investors · DB vs DC pensions · asset management industry
04	**The investment policy statement**	IPS 7 components · risk & return objectives · ability vs willingness · constraints (LTTLU) · Marie Gascon worked example
05	**Behavioral biases**	cognitive errors (fixable) vs emotional biases (adapt) · belief perseverance · processing · loss aversion · disposition effect
06	**Risk management framework**	risk governance · tolerance · budgeting · three lines of defense · closing the Deck 01 → Deck 20 arc

Learning objectives

By the end of this deck you should be able to:

1. Compute portfolio expected return and variance for a two-asset portfolio, and explain how correlation (not standard deviation alone) drives diversification benefits.

2. Derive an asset's expected return using the CAPM, distinguish systematic from idiosyncratic risk, and use the Security Market Line to identify under- and over-priced assets.

3. Describe the three-step portfolio management process and the role of the IPS — including how to resolve the conflict between a client's ability and willingness to take risk.

4. Distinguish cognitive errors from emotional biases and name the specific biases (anchoring, loss aversion, herding, overconfidence) most likely to impair portfolio decisions.

5. Explain the risk management framework, the difference between risk tolerance and risk budgeting, and connect the CAPM discount rate back to the DCF valuation framework of Deck 01.

SECTION 01

MPT foundations

Risk-averse investors require a higher expected return for every extra unit of variance they accept — utility encodes this trade-off numerically

Most investors are risk-averse: they demand compensation for bearing risk. The coefficient A is calibrated per-client in the IPS.

$$U = E(R) - \frac{1}{2} \times A \times \sigma^2$$

where: U = utility; A = coefficient of risk aversion; σ^2 = portfolio variance. Higher A → steeper indifference curves → more averse.

1. **A > 0 · Risk-averse**
   Typical investor. Demands higher return to accept variance. Indifference curves slope up and are convex — steeper as A rises. Retail clients: A ≈ 2–4; institutions: A ≈ 2.

2. **A = 0 · Risk-neutral**
   Indifferent between certain and uncertain outcomes of equal expected value. Theoretical benchmark — vanishingly rare in practice except in option-pricing math.

3. **A < 0 · Risk-seeking**
   Utility rises with variance. Gamblers, lottery-ticket buyers. Often exhibited in the loss domain per Kahneman–Tversky — risk-seeking when underwater.

Source: CFA Institute, Curriculum Vol 9, Learning Module 1, §4 "Risk Aversion and Portfolio Selection", pp. 12–20.

Portfolio return is a weighted average of asset returns — but portfolio variance depends on the correlation between assets, diversification's only free lunch

For a large N-asset portfolio, as N rises the w^2 variance terms vanish and average covariance dominates — the end-state of diversification.

$$E(R_p) = \Sigma\, w_i\, E(R_i) \qquad \sigma^2_p = \Sigma\, w_i^2\, \sigma_i^2 + \Sigma\, \Sigma_{i \neq j}\, w_i\, w_j\, Cov(R_i, R_j)$$

where: $Cov(R_i, R_j) = \rho_{ij} \times \sigma_i \times \sigma_j$. Return is linear in weights; risk is NOT — that asymmetry is what makes diversification work.

TWO-ASSET SPECIAL CASE

$\sigma^2_p = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2\, w_1\, w_2\, \rho\, \sigma_1\, \sigma_2$

This is the workhorse formula you memorize for CFA L1 — every multi-asset result extends from this base.

LARGE-N LIMIT

As $N \to \infty$ with equal weights, $\sigma^2_p \to$ average pairwise covariance. Individual-asset variance ceases to matter — only the covariance structure does.

Source: CFA Institute, Curriculum Vol 9, Learning Module 1, §7 "Portfolio Risk", pp. 21–25.

CFA LM 1 Example 3 — combining an 80% S&P 500 position with 20% Emerging Markets raises return and lowers risk at the same time

*Adding a *riskier* asset to a portfolio can REDUCE total portfolio risk when its correlation with the existing holdings is low enough — the classic MPT insight.*

Asset	Weight	E(R)	σ
S&P 500	80%	9.93%	16.21%
MSCI Emerging Mkts	20%	18.20%	33.11%
Covariance S&P / EM	–	–	0.0050
Portfolio (80/20)	100%	11.58%	15.10%

SOLUTION

Compute portfolio return (weighted avg) and variance using the two-asset formula.

1. $E(R_p) = 0.80 \times 9.93\% + 0.20 \times 18.20\% = 11.58\%$
2. $\sigma^2_p = (0.80^2 \times 0.1621^2) + (0.20^2 \times 0.3311^2) + (2 \times 0.80 \times 0.20 \times 0.0050)$
3. $= 0.01682 + 0.00439 + 0.00160 = 0.02281$
4. $\sigma_p = \sqrt{0.02281} = 15.10\%$

⇒ E(R)=11.58% and σ=15.10% — higher return AND lower risk than S&P alone.

Source: CFA Institute, Curriculum Vol 9, Learning Module 1, Example 3, pp. 22–24.

Correlation ρ is the lever: ρ = +1 erases diversification, ρ = −1 allows a risk-free portfolio, 0 ≤ ρ < 1 is where diversification actually lives

Two stocks with identical 10% return and 20% risk, combined 50/50. Only correlation changes. Portfolio variance collapses as ρ falls.

$$\sigma^2_p = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2\, w_1\, w_2\, \rho\, \sigma_1\, \sigma_2$$

Correlation ρ	Portfolio variance	Portfolio σ_p	Diversification benefit
+1.00	0.04 (= weighted avg)	20.00%	None — σ equals weighted avg
+0.50	0.030	17.32%	2.68 pp risk reduction
0.00	0.020	14.14%	5.86 pp — no co-movement
−0.50	0.010	10.00%	Half the risk erased
−1.00	0.000	0.00%	Perfect hedge — risk-free portfolio

KEY RULE

For any ρ < +1, portfolio standard deviation is LESS than the weighted average of individual standard deviations. Equality only at ρ = +1.

Source: CFA Institute, Curriculum Vol 9, Learning Module 1, §8 "Correlation and Risk Diversification", pp. 25–35.

The efficient frontier is the upper boundary of all portfolios with minimum variance at each expected return — every portfolio below it is dominated

Markowitz's 1952 insight: rational investors should hold only portfolios on the efficient frontier. Everything inside the opportunity set is sub-optimal.

THREE KEY PORTFOLIOS

Global Minimum-Variance Portfolio (GMVP): the single portfolio on the frontier with lowest possible risk.

Minimum-Variance Frontier: lowest σ for each level of E(R).

Efficient Frontier: the UPPER half of the minimum-variance frontier — from GMVP upward to the highest-return asset.

WHY THE LOWER HALF IS DOMINATED

Any portfolio on the lower half of the min-variance frontier has a mirror-image above it with the same σ but higher E(R).

No rational investor would ever choose the lower portfolio — hence "dominated."

Utility maximization selects a single point on the frontier where the highest indifference curve is tangent.

ECONOMIC MEANING

The efficient frontier is concave — doubling risk does NOT double return. Each marginal unit of σ buys less E(R) than the previous one.

SECTION 02

CAL, CML, CAPM & SML

Adding a risk-free asset produces the Capital Allocation Line — a straight line whose slope is the Sharpe ratio of the risky portfolio

The CAL dominates the efficient frontier of risky assets alone. Investors hold some combination of the risk-free asset and the optimal risky portfolio.

$$E(R_p) = R_f + [(E(R_i) - R_f) / \sigma_i] \times \sigma_p$$

where: Intercept = R_f; slope = $(E(R_i) - R_f) / \sigma_i$ = Sharpe ratio = "market price of risk". All portfolios on the CAL share this slope.

POINT R_f

100% in risk-free asset. $\sigma = 0$. Return = R_f. Leftmost point on CAL.

POINT i

100% in risky portfolio. $\sigma = \sigma_i$. Return = $E(R_i)$. Tangent point with efficient frontier — the *optimal* risky portfolio.

BEYOND i

Leverage: borrow at R_f, invest >100% in i. $w_{rf} < 0$; return rises linearly, σ rises linearly.

Source: CFA Institute, Curriculum Vol 9, Learning Module 2, §2 "Portfolio of Risk-Free and Risky Assets", pp. 64–68.

When every investor holds the same optimal risky portfolio — the market portfolio — the CAL becomes the Capital Market Line, and two-fund separation obtains

Under homogeneous expectations every investor picks the same risky portfolio = the value-weighted market portfolio M. Differences in risk tolerance only change the R_f/M mix.

$$E(R_p) = R_f + [(E(R_m) - R_f) / \sigma_m] \times \sigma_p$$

where: Slope = Sharpe ratio of the market = equity risk premium ÷ market σ. This is the CAL of the market portfolio — the CML.

TWO-FUND SEPARATION THEOREM

Every rational investor holds only two funds:
(1) the risk-free asset, and
(2) the market portfolio M.

Risk tolerance dictates the split; asset selection is the SAME for everyone — the core insight behind index investing.

PASSIVE VS ACTIVE

CML says passive = optimal if markets are efficient.

Active managers must either (a) beat the CML by skill ($\alpha > 0$) or (b) accept that they are below the CML with additional costs.

Source: CFA Institute, Curriculum Vol 9, Learning Module 2, §3 "The Capital Market Line", pp. 68–74.

Total risk = systematic + nonsystematic; only systematic risk is priced because diversification eliminates the rest for free

Idiosyncratic shocks wash out in a well-diversified portfolio ($N \gtrsim 30$ uncorrelated names). Market-wide shocks do not. The market pays only for the latter.

$$\sigma^2_{total} = \sigma^2_{systematic} + \sigma^2_{nonsystematic}$$

Risk type	Other names	Examples	Diversifiable ?
Systematic	market / non-diversifiable / β	Recession, policy rate move, pandemic, war	No
Nonsystematic	idiosyncratic / specific / α	CEO scandal, product recall, single-firm fraud	Yes

WHY ONLY SYSTEMATIC RISK IS COMPENSATED

If investors could earn a premium for idiosyncratic risk, they could diversify it away costlessly and still pocket the premium — an arbitrage the market eliminates. Thus in equilibrium, only non-diversifiable risk carries an expected-return premium.

Source: CFA Institute, Curriculum Vol 9, Learning Module 2, §5 "Systematic and Nonsystematic Risk", pp. 77–81.

Beta measures how much an asset moves with the market — the only risk the market compensates you for

β is estimated from historical data via regression of asset returns on market returns. Practitioners adjust raw β toward 1 (Blume adjustment) for forecasting.

$$\beta_i = Cov(R_i, R_m) / \sigma^2_m = \rho_{i,m} \times \sigma_i / \sigma_m$$

where: β is both the slope coefficient in $R_i = \alpha + \beta R_m + \varepsilon$ (market model) and the standardized measure of systematic risk.

Beta	Interpretation	Typical name
$\beta = 0$	Asset return uncorrelated with market. Return = R_f in CAPM.	Risk-free asset, Bajaj Auto (Ex. 7)
$0 < \beta < 1$	Less volatile than market. Defensive.	Utilities, consumer staples
$\beta = 1$	Moves one-for-one with the market by definition.	The market portfolio
		Tech.
PORTFOLIO BETA		
$\beta_p = \Sigma w_i \times \beta_i$ — a weighted average, exactly like portfolio return. Use this to target portfolio-level market exposure.		
$\beta < 0$	Moves opposite to market — rare, valuable hedges.	Gold (sometimes), VIX-linked products

The CAPM says expected return equals the risk-free rate plus beta times the equity risk premium — the single most-cited formula in modern finance

Rf is usually a government-bond yield; ERP is the long-run excess return of equities over R_f (typically 4–6% in developed markets).

$$E(R_i) = R_f + \beta_i \times [\, E(R_m) - R_f \,]$$

where: E(R_m) − R_f = equity risk premium (ERP). The CAPM linearly maps systematic risk (β) into required return.

CAPM WORKED EXAMPLE — COST OF EQUITY FOR A CYCLICAL STOCK

Stock has β = 1.2. Risk-free rate R_f = 2.0%. Equity risk premium E(R_m) − R_f = 5.0%.

1. Plug into CAPM: E(R_i) = 2.0% + 1.2 × 5.0%
2. Beta premium: 1.2 × 5.0% = 6.0%
3. Sum: 2.0% + 6.0% = 8.0%
4. This 8.0% is both the required return and the discount rate for DCF valuation of the stock's equity cash flows.

⇒ E(R) = 8.0% = required return = cost of equity = DCF discount rate.

[1] CAPM assumes: (i) homogeneous investor expectations, (ii) single holding period, (iii) investors are rational mean-variance optimizers, (iv) unlimited borrowing/lending at R_f.

Source: CFA Institute, CFA Curriculum, Vol 9, Learning Module 2, §7 "CAPM Assumptions and the Security Market Line", pp. 86–91.

CFA LM2 Example 7 — Bajaj Auto has zero correlation with the market, so its CAPM expected return collapses back to the risk-free rate

Assumptions shared by both companies: $R_f = 3\%$, $E(R_m) = 13\%$, $\sigma_m = 23\%$. Only correlation with the market and σ_i differ between the two.

BAJAJ AUTO (INDIA) · $\rho = 0.00$

$\sigma_i = 50\%$, $\rho_{i,m} = 0$ with the market. Compute β and $E(R)$.

1. $\beta = \rho \times \sigma_i / \sigma_m$
2. $= 0.00 \times 0.50 / 0.23 = 0.00$
3. $E(R) = 3\% + 0 \times (13\% - 3\%)$
4. $= 3.0\%$

⇒ $E(R) = R_f = 3.0\%$ — no systematic exposure = no premium.

MUELLER METALS (GERMANY) · $\rho = 0.65$

$\sigma_i = 50\%$, $\rho_{i,m} = 0.65$ with the market. Compute β and $E(R)$.

1. $\beta = \rho \times \sigma_i / \sigma_m$
2. $= 0.65 \times 0.50 / 0.23 = 1.41$
3. $E(R) = 3\% + 1.41 \times (13\% - 3\%)$
4. $= 3\% + 14.1\% = 17.1\%$

⇒ $E(R) = 17.1\%$ — high correlation drives the premium.

TAKEAWAY

Same total risk (σ=50%). Only the CORRELATION with the market differs — and that alone drives the 14pp return gap.

Source: CFA Institute, Curriculum Vol 9, Learning Module 2, Example 7, pp. 89–90.

The Security Market Line plots expected return against beta — assets above the line are underpriced, below the line are overpriced

Estimated return above CAPM required return ⇒ $\alpha > 0$ ⇒ BUY. Below ⇒ $\alpha < 0$ ⇒ SELL. In equilibrium $\alpha = 0$ for all assets and every point sits ON the SML.

ABOVE SML · UNDERPRICED

Estimated return > CAPM required return for the given β.

$\alpha > 0$.

Action: BUY — the market is paying you MORE than its equilibrium compensation for the risk taken.

ON THE SML · FAIRLY PRICED

Estimated return = CAPM required return for the given β.

$\alpha = 0$.

Action: HOLD at market weight. In CAPM equilibrium, every asset sits exactly on the SML — no free lunch.

BELOW SML · OVERPRICED

Estimated return < CAPM required return for the given β.

$\alpha < 0$.

Action: SELL / UNDERWEIGHT — compensation is inadequate for the β exposure. Long-only funds underweight; hedge funds short.

[1] The SML differs from the CML: CML uses total risk (σ) and applies only to efficient portfolios. SML uses systematic risk (β) and applies to ANY asset, efficient or not.

Source: CFA Institute, Curriculum Vol 9, Learning Module 2, §7 "The Security Market Line", pp. 88–91.

SECTION 03

The portfolio process

Every professional portfolio is built in the same three steps — plan, execute, feed back — and the IPS anchors all three

This is the single most-tested workflow on CFA L1 portfolio management. Sequence matters: skipping the planning step is the #1 practitioner failure.

1

STEP 1 · PLANNING

Understand the client (fact-finding, questionnaires, interviews) → draft the IPS (investment policy statement). Output: written IPS with risk/return objectives and constraints. Reviewed every 3 years or at any material change.

2

STEP 2 · EXECUTION

Strategic asset allocation (SAA) → security analysis (top-down + bottom-up) → portfolio construction (sizing, sector tilts, trading). The SAA is the single biggest driver of long-run portfolio returns.

3

STEP 3 · FEEDBACK

Monitor portfolio → rebalance when drift > threshold → evaluate performance vs benchmark → report to client. Feedback loop may trigger IPS revision if circumstances have changed.

Source: CFA Institute, Curriculum Vol 9, Learning Module 3, §4 "Steps in the Portfolio Management Process", pp. 133–137.

Individual vs institutional investors face very different horizons, liabilities, and constraints — one portfolio template does not fit all

Yale endowment: 80%+ alternatives / equities, <5% fixed income. MassMutual insurer: 80%+ bonds. Same economy, opposite portfolios — because their liabilities differ.

Investor type	Horizon	Risk tol.	Liquidity	Primary constraint
Retail individual	10–50y	Variable	Moderate	Taxes; unique circumstances
DB pension plan	Long (LDI)	Low-moderate	Low	Liabilities + regulation
DC pension plan	Participant's	Participant's	High at retire	Participant chooses
Endowment / foundation	Perpetual	High	Low	Spending policy (~5%)
Insurer (life / P&C)	Matched	Low	High for P&C	Regulation (Solvency II etc.)
Sovereign wealth fund	Very long	High	Low	Political mandate

Source: CFA Institute, Curriculum Vol 9, Learning Module 3, §5 "Types of Investors", pp. 137–143.

DB and DC pensions place the investment risk on opposite sides of the ledger — the key distinction every portfolio manager must internalize

If the Deck 01 discount-rate framework feels abstract, a DB plan makes it concrete: the plan's PV of liabilities IS a DCF calculation, and funded ratio = assets / DCF.

DEFINED BENEFIT (DB)

Employer promises a specified pension formula (e.g. 2% × years × final salary).

INVESTMENT RISK: on the EMPLOYER.

LONGEVITY RISK: on the EMPLOYER.

Asset-liability management (LDI) framework. Discount liabilities at a bond-like rate; match duration with bonds. Underfunded DB plans are off-balance-sheet corporate debt.

DEFINED CONTRIBUTION (DC)

Employer contributes a specified amount (e.g. 5% match on 5% employee) — not a specified payout.

INVESTMENT RISK: on the EMPLOYEE.

LONGEVITY RISK: on the EMPLOYEE.

Individual 401(k) / personal pension accounts. Target-date / lifestyle funds are the default. Employee bears all capital-market outcomes.

THE GLOBAL SHIFT

Since the 1990s, most private-sector plans have shifted from DB to DC — transferring decades of investment and longevity risk from corporate balance sheets to households.

Source: CFA Institute, Curriculum Vol 9, Learning Module 3, §5.1 "Defined Benefit Pension Plans", pp. 138–143.

SECTION 04

The IPS

Every IPS answers two questions in turn: what are the client's risk and return objectives, and what constraints apply to meeting them

The IPS is a legal requirement in many jurisdictions (UK Pensions Act 1995, EU MiFID KYC) — and a defensive document for the adviser when clients push back later.

Section	Contents
1. Introduction	Describes the client (individual, institution). Basic demographics, wealth level, sources of income.
2. Statement of Purpose	Why the IPS exists. Goals the portfolio is to achieve. Framework for measuring success.
3. Duties and Responsibilities	Of client, custodian, investment manager, any sub-advisers. Fiduciary obligations.
4. Procedures	Steps to keep the IPS current. Process for responding to market events, rebalancing, contingencies.
5. Investment Objectives	Return and risk objectives, absolute or relative to benchmarks. Must be consistent with each other.
6. Investment Constraints	Liquidity, time horizon, tax, legal/regulatory, unique (the "LTTLU" mnemonic). ESG if applicable.
7. Investment Guidelines	Permissible asset classes, use of leverage, derivatives, excluded securities. + SAA appendix (policy portfolio).

Source: CFA Institute, Curriculum Vol 9, Learning Module 4, §2 "The Investment Policy Statement", pp. 162–164.

Risk tolerance is the lower of ability (objective) and willingness (subjective) — never average them; always defer to the more conservative

If ability and willingness conflict, the adviser counsels the client, documents the gap, and SETS the portfolio at the lower of the two. Never modify personality.

Risk tolerance = MIN (Ability to bear risk , Willingness to take risk)

	Willingness: BELOW avg	Willingness: ABOVE avg
Ability: ABOVE avg	Resolution needed — educate, document gap, default to BELOW avg	CONSISTENT → above-average tolerance
Ability: BELOW avg	CONSISTENT → below-average tolerance	Resolution needed — counsel, default to BELOW avg

ABILITY = OBJECTIVE, WILLINGNESS = SUBJECTIVE

ABILITY drivers (measurable): time horizon, wealth/liabilities ratio, income stability, expected future expenses. LONGER horizon + HIGHER wealth = higher ability.

WILLINGNESS drivers (psychological): risk attitude questionnaires, past investment experience, self-reported tolerance. Not the adviser's job to change it.

Source: CFA Institute, Curriculum Vol 9, Learning Module 4, §3 "IPS Risk and Return Objectives", pp. 164–170.

The "RR LTTLU" template covers every IPS constraint — Liquidity, Time horizon, Tax, Legal/regulatory, Unique, on top of Risk & Return objectives

Worked for a DC participant: Sophie, 45yo, £80k salary, 20y to retirement, targeting £700k retirement pot. Fill each row to produce a complete IPS.

Component	Definition	Sophie's IPS entry
RETURN objective	Absolute or relative; nominal or real; before or after fees/tax	Real 4.5% p.a. over 20y (nominal ~7% @ 2.5% inflation)
RISK objective	Absolute (σ, max drawdown) or relative (tracking error)	Max 15% annual σ; drawdown limit 25% in any 12m
LIQUIDITY	Upcoming cash needs from the portfolio	Emergency fund £20k; no planned drawdowns before age 65
TIME horizon	Single- or multi-stage; may include accumulation + decumulation	Two-stage: 20y accumulation; 25y retirement decumulation
TAXES	Marginal rates; tax-advantaged wrappers (pensions, ISAs)	UK higher-rate 40%; max pension contribution relief
LEGAL / regulatory	Jurisdictional rules, trust deeds, corporate restrictions	UK DC rules; 25% tax-free lump sum from age 55
UNIQUE circumstances	ESG preferences, concentrated holdings, illiquid assets	Employer-stock holdings must not exceed 10%

CFA LM4 Example 4 — Marie Gascon's case solves the annual return she needs to fund a €2m retirement in 20 years at 2% inflation

The return objective flows straight from the constraint analysis (inflation + horizon + capital) — NOT from subjective market expectations. TVM math from Deck 01.

MARIE GASCON · AGE 30 · RETIRE AT 50 · 20-YEAR HORIZON

Current savings €1m total. Earmark €100k as emergency cash, invest €900k. Target: €2m in today's money at retirement. Inflation expected 2% p.a. Pension contributions and returns are tax-exempt in France; distributions at retirement taxable.

1. Step 1 — Inflate the target: €2,000,000 × (1.02)^20 = €2,971,895 (nominal money-of-the-day)
2. Step 2 — Required CAGR on €900k to reach €2,971,895: (€2,971,895 / €900,000)^(1/20) − 1
3. Step 3 — Ratio = 3.302; 3.302^(1/20) = 1.0617 → 6.17% nominal p.a.
4. Step 4 — Real required return: (1.0617 / 1.02) − 1 = 4.09% real p.a.
5. Step 5 — Verify: €900,000 × (1.062)^20 ≈ €2,997,000 ≈ target. ✓

⇒ Required nominal return 6.2% p.a. (4.1% real) — now build a portfolio that plausibly delivers it within Marie's risk tolerance.

Source: CFA Institute, Curriculum Vol 9, Learning Module 4, Example 4, pp. 169–170.

SECTION 05

Behavioral biases

Behavioral biases split cleanly into cognitive errors — fixable with better information — and emotional biases, which you can only recognize and adapt around

The distinction drives the remediation: for cognitive biases → more / better data, education, decision checklists. For emotional biases → structural portfolio rules the client cannot override.

COGNITIVE ERRORS

SOURCE: faulty reasoning. Mistakes made while processing information.

SPLIT: belief-perseverance (conservatism, confirmation, representativeness, illusion of control, hindsight) and processing (anchoring, mental accounting, framing, availability).

FIX: yes — with better information, education, advice, systematic decision frameworks.

EMOTIONAL BIASES

SOURCE: impulses, intuitions, feelings. Not conscious errors — they arise spontaneously.

SIX: loss aversion, overconfidence, self-control, status quo, endowment, regret aversion.

FIX: no — recognize and ADAPT. Structural portfolio rules (rebalancing thresholds, sell-discipline, DCA) that remove discretion from the client.

RULE OF THUMB

Cognitive → educate. Emotional → automate. If you can't "teach your way out" of the bias, engineer it out of the decision path.

Source: CFA Institute, Curriculum Vol 9, Learning Module 5, §2 "Behavioral Bias Categories", pp. 206–207.

Fifteen biases fit a 2×2 of source × mechanism — anchoring and loss aversion are the two that move markets most

This is the single diagram every CFA L1 candidate should memorize. Count: 5 belief-perseverance + 4 processing + 6 emotional = 15 biases.

Category	Bias	Quick definition
Cognitive — belief persev.	Conservatism	Underweight new info vs prior beliefs
Cognitive — belief persev.	Confirmation	Seek info that confirms; ignore contradicting
Cognitive — belief persev.	Representativeness	Classify on superficial similarity; ignore base rates
Cognitive — belief persev.	Illusion of control	Overestimate personal influence on outcomes
Cognitive — belief persev.	Hindsight	"I knew it all along" after the fact
Cognitive — processing	Anchoring & adjustment	Fix estimates too close to initial (e.g. purchase) value
Cognitive — processing	Mental accounting	Treat fungible money as separate "pots"
Cognitive — processing	Framing	Answer differently based on presentation (gain vs loss)
Cognitive — processing	Availability	Weight recent / memorable events too heavily
Emotional	Loss aversion	Loss ≈ 2× as painful as equivalent gain; disposition effect

Loss aversion — a loss hurts roughly 2.25× as much as an equivalent gain feels good — is the emotional bias with the biggest portfolio consequences

Kahneman & Tversky (1979): utility function is S-shaped around a reference point — concave in gains (risk-averse) and CONVEX in losses (risk-SEEKING).

THE S-SHAPED VALUE FUNCTION

Upper half (gains): concave → risk-averse. A 100→110 gain feels less than 50% as good as a 100→120 gain.

Lower half (losses): convex → risk-seeking. Investor will "double down" on losing positions to avoid realizing the loss.

Key property: the function is ~2.25× steeper on the loss side than the gain side.

THE DISPOSITION EFFECT

Loss aversion → hold losers too long (hoping for breakeven) and sell winners too soon (to lock in gains).

Empirical result: loss-averse investors underperform passive benchmarks by ~1–2% p.a. on average.

FIX: structural stop-losses, pre-committed rebalancing rules, tax-loss harvesting to remove the emotional "lock-in" of losses.

PM IMPLICATION

Automate sell-discipline. Human discretion at the exit point is where loss aversion destroys the most value — pre-commit mechanisms bypass it.

Source: CFA Institute, Curriculum Vol 9, Learning Module 5, §4 "Loss-Aversion Bias", pp. 219–221.

SECTION 06

Risk management

Risk management is not risk minimization — it's choosing and sizing risks to fit the enterprise's tolerance, then monitoring continuously

"The Doctrine of No Surprises" — a good risk manager is not expected to predict events; they are expected to have quantified and stress-tested the impact of those events in advance.

Framework element	What it does
1. Risk governance	Board-level: defines goals, risk appetite, oversight, delegation. Top-down.
2. Risk identification & measurement	Qualitative + quantitative inventory of exposures. σ, VaR, stress tests, scenarios.
3. Risk infrastructure	People + systems: data, models, engines, reports. Scale depends on complexity.
4. Policies & processes	Limits, procedures, checklists. Operationalizes governance into daily actions.
5. Risk monitoring, mitigation, mgmt.	Continuous. When exposure breaches tolerance: mitigate (prevent, avoid, accept, transfer).
6. Communications	Timely reports across all levels; feedback loops with governance body.
7. Strategic analysis / integration	Risk management as OFFENSIVE tool — improves risk-adjusted decision quality.

Source: CFA Institute, Curriculum Vol 9, Learning Module 6, §2–3 "Risk Management Process & Framework", pp. 243–250.

Risk tolerance and risk budgeting bridge the gap between the board's appetite and the PM's daily trading decisions

Tolerance is the appetite statement ("not more than X% loss in a year"); budgeting allocates that X% across strategies, assets, and factors in a measurable way.

RISK TOLERANCE

WHAT risks are acceptable, and at what level?

Set by the BOARD. Informed by two analyses:

- Inside view: what failures we cannot tolerate
- Outside view: what drivers we're exposed to

RISK BUDGETING

HOW MUCH of the tolerance is spent on each strategy / asset / factor?

Common metrics: σ, β, VaR, scenario loss.

Implements tolerance operationally.

1ST LINE · BUSINESS

PMs, traders, front office. Own the risks. Make the decisions.

2ND LINE · RISK & COMPLIANCE

Risk team, CRO, compliance. Independent oversight of 1st-line.

3RD LINE · INTERNAL AUDIT

Periodic review of both 1st and 2nd lines. Reports to board.

Source: CFA Institute, Curriculum Vol 9, Learning Module 6, §5–6 "Risk Tolerance" and "Risk Budgeting", pp. 254–259.

The full arc of Deck 01 (TVM) to Deck 20 (MPT/CAPM) collapses into a single statement: the discount rate we started with is the equilibrium price of risk

The "r" in every DCF from Deck 01 onwards is no longer an exogenous assumption. MPT/CAPM tells us exactly what it should be in equilibrium — and why.

$$PV = \Sigma\ CF_t / (1 + r)^t \qquad \text{where} \quad r = R_f + \beta \times [\ E(R_m) - R_f\ ]$$

where: Deck 01 formula (left) meets Deck 20 formula (right). The course's two bookends are the same equation viewed from two sides.

Deck	Concept	How it contributes to r or DCF
01	TVM & DCF	Defined PV, FV, NPV, IRR; treated r as given
02–08	Financial statements & ratios	Supplied the cash flows (CF_t) used in the numerator
09–13	Equity & fixed-income	Built payout forecasts; introduced yield spreads & ERP
14–17	Quant methods & portfolios	Showed how correlations shape portfolio σ
18–19	Alternatives & risk tools	Added non-systematic premia (illiquidity, complexity)
20	MPT & CAPM (this deck)	Derived r as R_f + β·ERP — the missing piece

Source: Synthesis — CFA Level I Volumes 1–9; ties back to Deck 01 §1 slides 5–8.

KEY TAKEAWAYS

Deck 20 — Portfolio Management, MPT, CAPM & Risk Management

- Portfolio return is a weighted average of asset returns; portfolio variance depends on correlation — diversification's one free lunch lives at $\rho < +1$.
- Efficient frontier = upper half of min-var frontier. CAL adds the risk-free asset; under homogeneous expectations it becomes the Capital Market Line (two-fund separation).
- Only systematic (β) risk is priced. CAPM: $E(R) = R_f + \beta \times ERP$. SML positions any asset — above = underpriced ($\alpha>0$ buy), below = overpriced ($\alpha<0$ sell).
- Portfolio process = Plan → Execute → Feed back. IPS anchors all three; its 7 sections separate objectives (risk/return) from constraints (LTTLU).
- Risk tolerance = MIN(ability, willingness). Ability is objective (horizon, wealth); willingness is subjective (attitude). Never average — always default to lower.
- Biases split into cognitive (fixable via info/education) and emotional (adapt via structural rules). Loss aversion is the single most portfolio-damaging bias.
- Risk management ≠ minimization. Framework = governance + measurement + infrastructure + policies + monitoring + comms + strategy. Budget the tolerance; 3 lines of defense.
- Deck 01 → Deck 20 arc: the "r" in every DCF is the CAPM equilibrium cost of capital — $R_f + \beta \cdot ERP$. 20 decks, one unifying equation. Thank you for the journey.